HACETTEPE UNIVERSITY INSTITUTE OF POPULATION STUDIES (HUIPS)

# DEM 601 Introduction to Demographic Analysis

**Formula sheet** COVERS lectures 1 to 5 TERM Fall 2026

> **Every equation the course has shown you, in the order it was shown.**

## Lecture 1 · Introduction to the Course and Basic Demographic Concepts

$$P_{n+k} = P_n + \underbrace{\underbrace{{}_kB_n - {}_kD_n}_{\text{natural increase}} + \underbrace{{}_kI_n - {}_kE_n}_{\text{net migration}}}_{\text{population change}}$$

What makes a population change

## Lecture 2 · Demographic Indicators and Person-Years-Lived

$$\text{indicator} = \frac{\text{numerator}}{\text{denominator}} \times k$$

What an indicator is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

The mean

$$M = L_0 + \left(\frac{P/2 - \sum f_0}{f_0}\right) i$$

The median of grouped data

$$\text{ratio} = \frac{a}{b} \times k$$

Ratio

$$\text{proportion} = \frac{a}{a+b}$$

Proportion

$$\text{percentage} = \frac{a}{a+b} \times 100$$

Percentage

$$\text{rate} = \frac{e}{PY} \times k$$

A rate needs time

$$CBR = \frac{\text{births in the year}}{\text{mid-year population}} \times 1000$$

Is it a true rate?

$$IMR = \frac{\text{deaths under age 1}}{\text{live births in the year}} \times 1000$$

Is it a true rate?

$$q = \frac{e}{P_0}$$

A rate is not a probability

$$\bar{P} = \frac{P_0 + P_1}{2}$$

The mid-year population

## Lecture 3 · Population Change

$$\Delta P = P_t - P_0$$

Absolute change

$$\%\Delta P = \frac{P_t - P_0}{P_0} \times 100$$

Percentage change

$$P_n = P_0 + \big(n \times (r \times P_0)\big)$$

Arithmetic growth

$$r = \frac{P_n - P_0}{n} \div P_0 \times 100$$

Arithmetic growth

$$P_n = P_0 \times (1 + r)^n$$

Geometric growth

$$r = \sqrt[n]{\frac{P_n}{P_0}} - 1$$

Geometric growth

$$P_n = P_0 \times e^{r \times n}$$

Exponential growth

$$r = \frac{\ln(P_n / P_0)}{n}$$

Exponential growth

$$r(t) = \lim_{\Delta t \to 0} \frac{\Delta N(t)}{N(t)\,\Delta t} = \frac{\frac{dN(t)}{dt}}{N(t)} = \frac{d\ln[N(t)]}{dt}$$

Exponential growth

$$P_n = P_0 \times e^{r \times n}$$

Running the formula forwards

$$2P_0 = P_0 \times e^{r \times n}$$

How long until twice as many?

$$\ln(2) = r \times n$$

How long until twice as many?

$$n = \frac{\ln(2)}{r}$$

How long until twice as many?

$$\text{Population Turnover} = \text{Losses} + \text{Gross Growth}$$

Standing still is work

## Lecture 4 · Age and Sex Composition

$$SR = \frac{P^m}{P^f} \times 100$$

Men per hundred women

$$SR_x = \underbrace{SRB}_{\text{where the cohort began}} \times \underbrace{\frac{S_x^m}{S_x^f}}_{\text{surviving to age } x} \times \underbrace{M_x}_{\text{net migration}}$$

Three things move it

$$\text{Child} = \frac{P_{\text{0-14}}}{P_{\text{15-64}}} \times 100$$

Four ratios from three numbers

$$\text{Elderly} = \frac{P_{65+}}{P_{\text{15-64}}} \times 100$$

Four ratios from three numbers

$$\text{General} = \text{Child} + \text{Elderly}$$

Four ratios from three numbers

$$\text{Aging index} = \frac{P_{65+}}{P_{\text{0-14}}} \times 100$$

Four ratios from three numbers

$$f = \frac{\text{total}}{\text{of known age}}$$

Putting back the ages nobody wrote down

$$W = 100 \times \frac{5\left(P_{25} + P_{30} + \cdots + P_{60}\right)}{P_{23} + P_{24} + \cdots + P_{62}}$$

Whipple's index

$$ID = \tfrac{1}{2}\sum_i \left|a_i - b_i\right|$$

The index of dissimilarity

## Lecture 5 · Measures of Mortality

$$CDR = \frac{D}{PY} \times 1000$$

Deaths per thousand people

$$ {}_{n}m_{x} \;=\; \frac{{}_{n}D_{x}}{{}_{n}PY_{x}} \times 1000 $$

One age at a time

$$ q_{x} \;=\; \frac{2m_{x}}{2 + m_{x}} \qquad\qquad {}_{n}q_{x} \;=\; \frac{2n \cdot {}_{n}m_{x}}{2 + n \cdot {}_{n}m_{x}} $$

From one to the other

$$ \text{share}_{c} \;=\; \frac{D_{c}}{D} \times 100 \qquad\qquad \text{rate}_{c} \;=\; \frac{D_{c}}{PY} \times 100{,}000 $$

A share is not a risk